

Real-Time Sales Leaderboard System - Summary

Goal: Build a real-time leaderboard platform for sales with low-latency updates, multi-tenant isolation, and strong correctness.

Key Flows:

1. Ingestion: Sales events flow from POS → API Gateway → Kafka (partitioned by tenant/branch).
2. Stream Processing: Flink/Kafka Streams normalize currency, deduplicate, aggregate sales.
3. Hot Storage: Redis ZSETs maintain leaderboards; metadata stored in hashes.
4. System of Record: Postgres/DynamoDB ensures durability and replay support.
5. Analytics: ClickHouse/Snowflake support historical and analytical queries.
6. Serving APIs:
 - POST /v1/event → ingest sales event
 - GET /v1/leaderboard → top N sellers
 - GET /v1/rank → rank for a seller
 - GET /v1/around → slice around a seller
 - WebSockets for delta updates/notifications

Performance Targets:

- Ingestion-to-visibility: $\leq 2s$ (p95).
- Query latency: $< 10ms$ (p95).
- Redis ops: $\sim 3ms$, end-to-end query $< 10ms$.

Security & Compliance:

- TLS 1.3, KMS-encrypted storage.
- Pseudonymized IDs, GDPR/CCPA compliant.
- Role-based and attribute-based access control.

Phased Build:

- MVP: Single region, Redis + Postgres/Dynamo.
- Phase 2: Multi-region, FX normalization, fraud flags.
- Phase 3: Global plane, ML-based anomaly detection.

End-to-End Flow:

Sale → API Gateway → Kafka → Flink → Redis + Analytics → Metadata Store → Leaderboards → APIs/WebSockets → Client devices.